



Doğa Yılmaz

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EDUCATION

2024 – Present

PhD in Computer Science – University College London

- Faculty of Engineering Sciences, Department of Computer Science
- Advisor: Assoc. Prof. Dr. He Wang and Prof. Tobias Ritschel
- Research Interests: Computer Graphics, Physics Informed Neural Rendering

2021 – 2023

MSc in Artificial Intelligence – Özyeğin University

- Faculty of Engineering, Department of Computer Science – Awarded Fellowship – GPA: 3.81/4.00
- Advisor: Assoc. Prof. Dr. Furkan Kırar
- Research Interests: Computer Vision, Computer Graphics, Inverse Rendering
- Thesis: Illumination-Guided Inverse Rendering Benchmark: Learning Real Objects with Few Cameras

2016 – 2020

BSc in Computer Science – Özyeğin University

- Faculty of Engineering, Department of Computer Science
- Advisor: Assoc. Prof. Dr. Furkan Kırar
- Thesis: Deep Residual Autoencoder for Real Image Denoising

PUBLICATIONS

- 2024 Yılmaz D., Takikawa T., Ceylan D. and Akşit, K., Learned Single-Pass Multitasking Perceptual Graphics for Immersive Displays, arXiv preprint arXiv:2408.07836
- 2023 Kınlı F., Yılmaz D., Özcan B., and Kırar F., DeNIM: Deterministic Neural Illuminant Mapping for Efficient Auto-White Balance Correction, IEEE ICCV Workshop on Resource Efficient Deep Learning for Computer Vision, 2023.
- 2023 Yılmaz D. and Kırar F., Illumination-guided inverse rendering benchmark: Learning real objects with few cameras. Computers & Graphics, 115, 107-121.
- 2023 Kınlı F., Yılmaz D., Özcan B., and Kırar F., Modeling the Lighting in Scenes as Style for Auto White-Balance Correction, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2023.
- 2022 Yılmaz, D., Kınlı, F., Özcan, B., and Kırar, F., "[Re] Lifting 2D StyleGAN for 3D-Aware Face Generation", ReScience C, 8(2), 2022. Presented at NeurIPS 2022 Journal Track.

EXPERIENCE

08/2022 – 01/2025

Computer Vision Research Engineer

Fishency Innovation – Stavanger, Norway (Remote)

- Key contributor to the development of an inverse rendering-based state-of-the-art fish biomass estimation project, leading to a peer-reviewed and published algorithm, and a highly accurate system with over 90% accuracy in biomass estimation. Employed software tools, including Mitsuba 3, PyTorch, and Python.
- Deployed and scaled the underwater fish biomass estimation software to the existing hybrid infrastructure.
- Developed 2D/3D visualization tools to validate and debug several machine learning pipelines.

11/2023 – 10/2024

Research Intern

University College London Computational Light Laboratory (CLL) – London, UK

- Conducted research in perceptual graphics and vision under Dr. Kaan Akşit's mentorship, in collaboration with Adobe Research.
- Contributed to the development of "Odak," an open-source scientific computing framework for optical sciences, computer graphics, and visual perception.

02/2021 – 10/2023

Graduate Research & Teaching Assistant

Özyeğin University Vision and Graphics Laboratory (VGL) – Istanbul, Turkey

- Actively participated in research mainly on inverse rendering and auto-white balance correction, yielding multiple publications.
- Contributed as a teaching assistant across a range of university courses, offering academic support and guidance to students.
- Courses Assisted: Advanced C++ Programming, Programming Paradigms, Data Structures and Algorithms, Object-Oriented Programming, Agile Software Development
- Academic Service: Reviewer, RCV Workshop ICCV; Reviewer, ReScience C Journal

AWARDS AND ACHIEVEMENTS

02/2024

Received a financial award from the Scientific and Technological Research Council of Turkey¹ for international publications.

10/2020

Ranked 1st in Turkey and 172nd globally out of 2155 teams in IEEEExtreme² 14 programming competition.

SKILLS

Languages

English: Advanced (IELTS: 7.5/9), German: Beginner, Turkish: Native

Programming

Python, Modern C++(17/20), Java

Technologies

PyTorch/LibTorch, Mitsuba 3, Kornia, OpenCV, OpenMP, AWS(S3, EC2, Rekognition), Docker, Blender3D

EXTRACURRICULAR ACTIVITIES

2020

Led the organization of Global Game Jam (GGJ)³ 2020 at Özyeğin University.

2019

Coordinated the IEEE Computer Society Özyeğin University Student Branch in the 2019 academic year.

¹The Scientific and Technological Research Council of Turkey (TÜBİTAK) serves as the governmental agency responsible for the scientific and technological affairs of Turkey.

²IEEEExtreme is a global challenge in which teams compete in a 24-hour time span against each other to solve a set of programming problems.

³Global Game Jam® (GGJ) is the world's largest game jam (game creation) event taking place around the world.